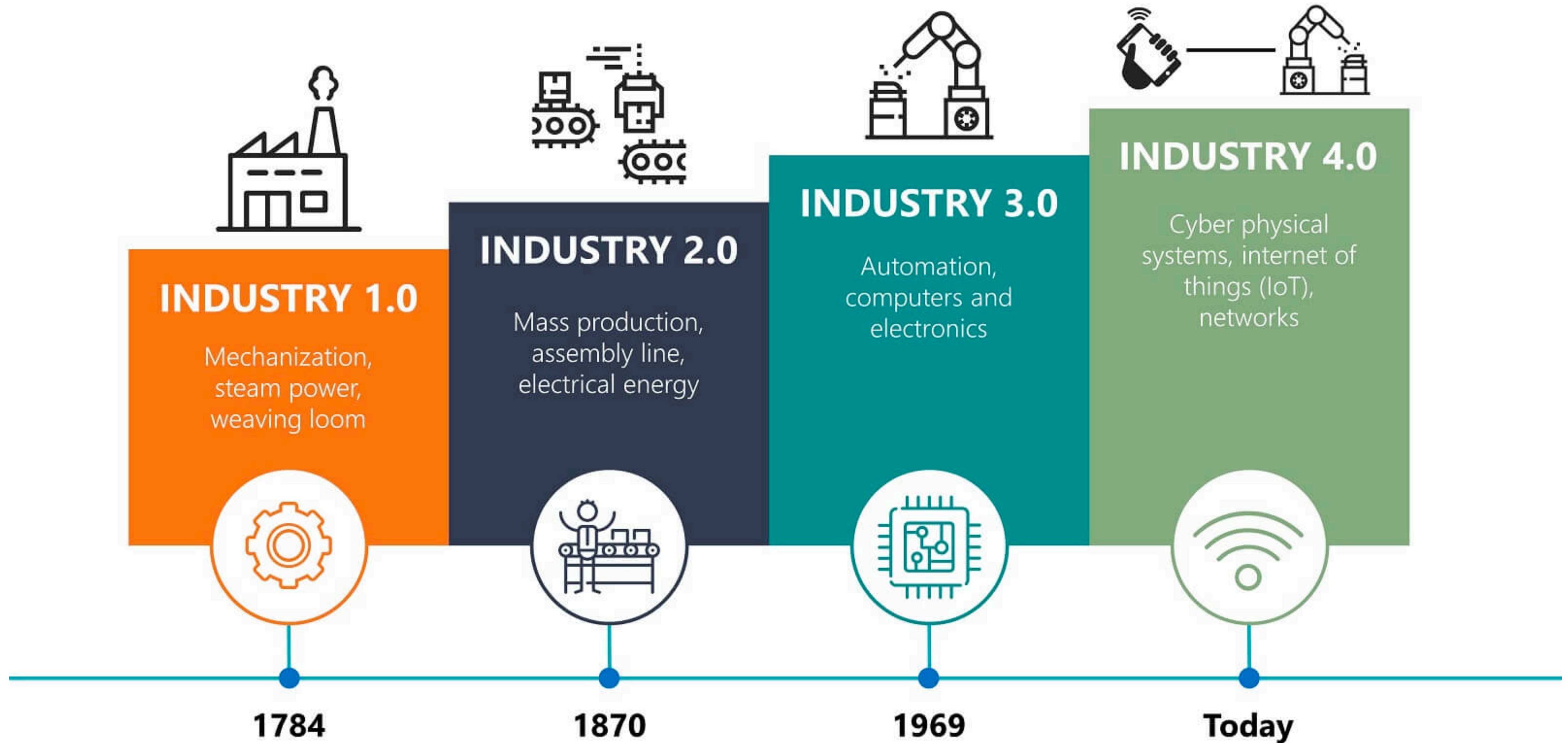


Software Dependability

IoT architectures

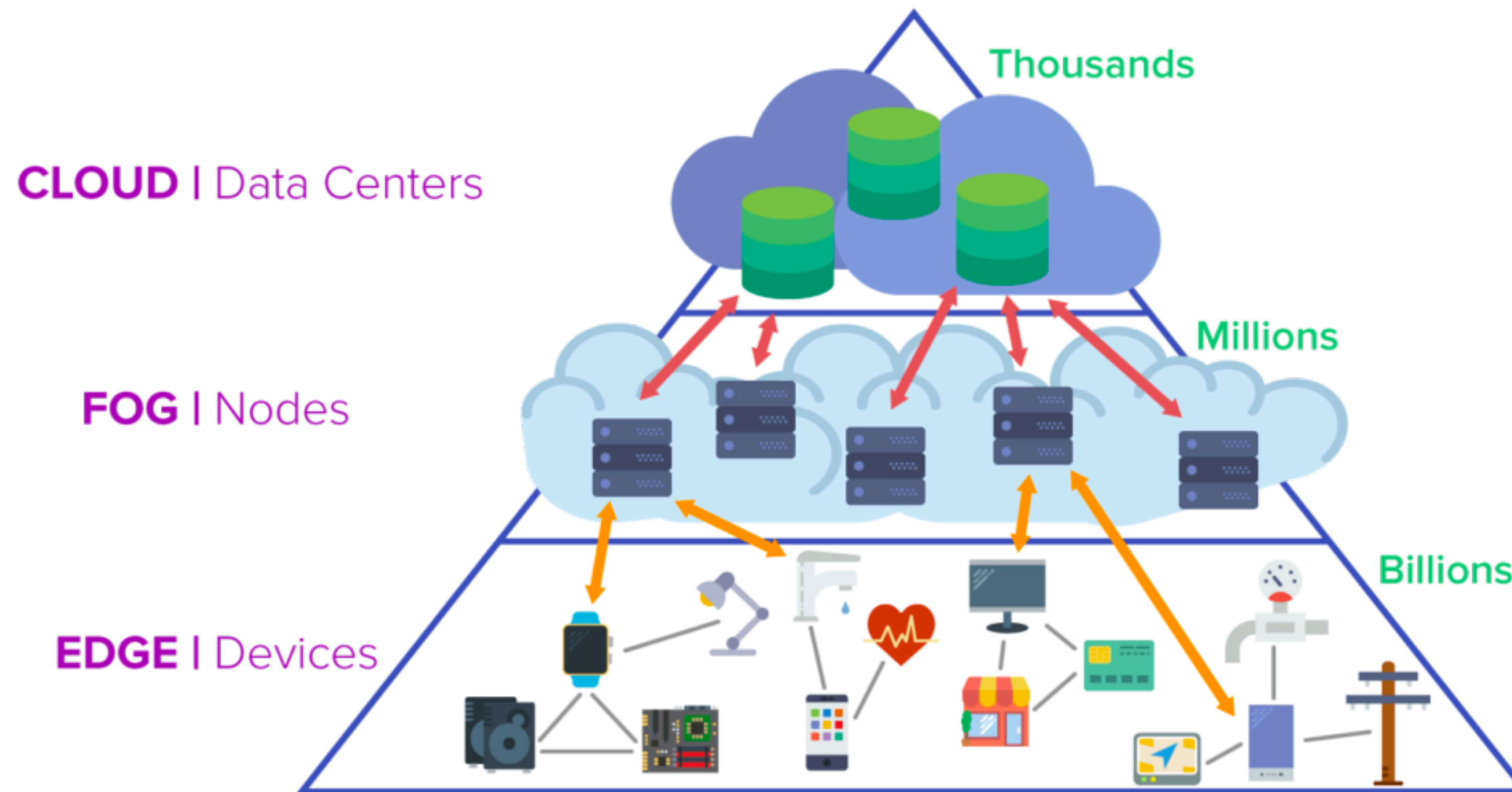
Industry 4.0?



Internet of Things VS. Industrial Internet of Things

IoT	VS	IIoT
Commercial or Consumer Convenience	Area of Focus	Monitoring and Managing Systems for High-stake Industries- Defense, Manufacturing, Health care & Others
Smart Devices	Focus Development	Sophisticated Machines
Sensitive Sensors, Advanced Controls and Analytics	Degree of Application	Simple Application with Low-risk Impacts
Utility-centric	Security and Risk Measures	Advanced and Robust
Functionally Independent	Interoperability	Integration with Co-existing Legacy Operations Systems
Low Scale Networks	Scalability	Large Scale Networks
Critically Monitored	Precision and Accuracy	Synchronized to Milliseconds
Easy Off-site Programming	Programmability	Remote on-site Reprogramming Required to Support New Processes
Convenience	Output	Economic Growth
Not Required	Resilience	Must be Automated to Support Fault Tolerance
Consumer Preferred	Maintenance	Scheduled and Organized

Software Architecture to support IoT



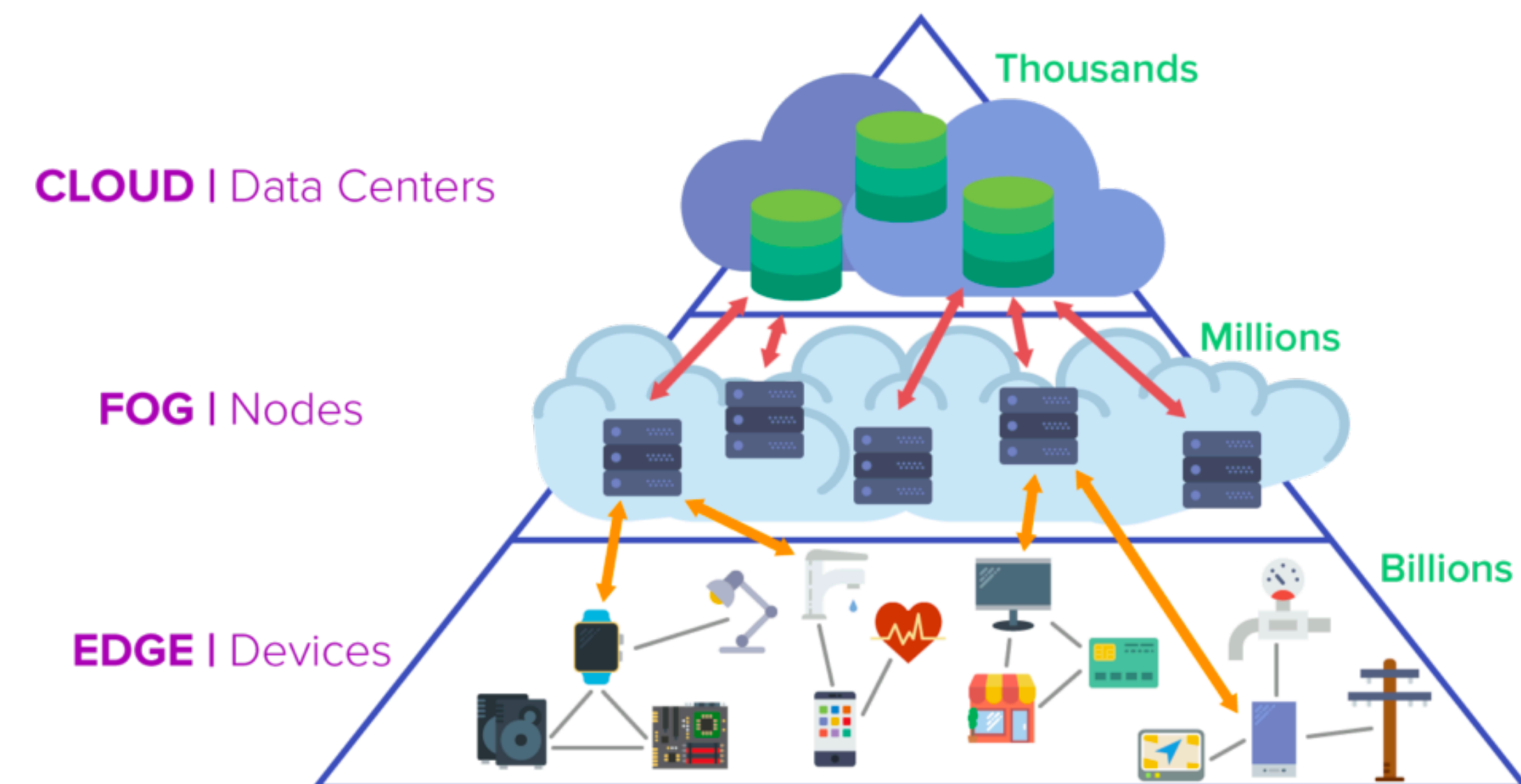
Software Architecture to support IoT

Cloud-centric architectures

Data from IoT devices (e.g., sensors) are transmitted in streaming to data centers where all data-intensive applications are executed. Cloud servers control the edge devices.

Device-centric architectures (i.e., Fog and Edge computing)

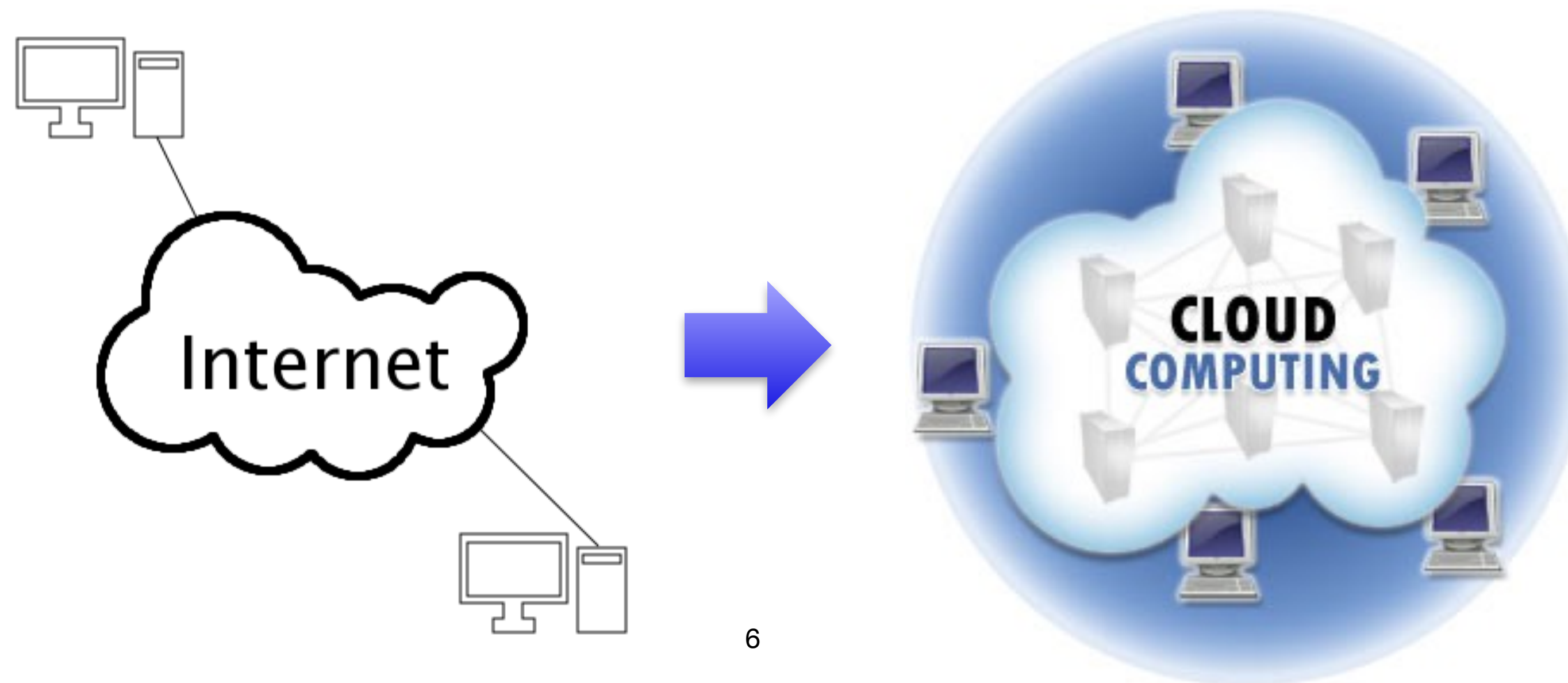
All data are elaborated in the devices with minimal interaction with the cloud for updated firmware or provisioning.



What is Cloud Computing?

Cloud computing is a general term for anything that involves **delivering hosted services over the internet**.

whatis.com



What is Cloud Computing?

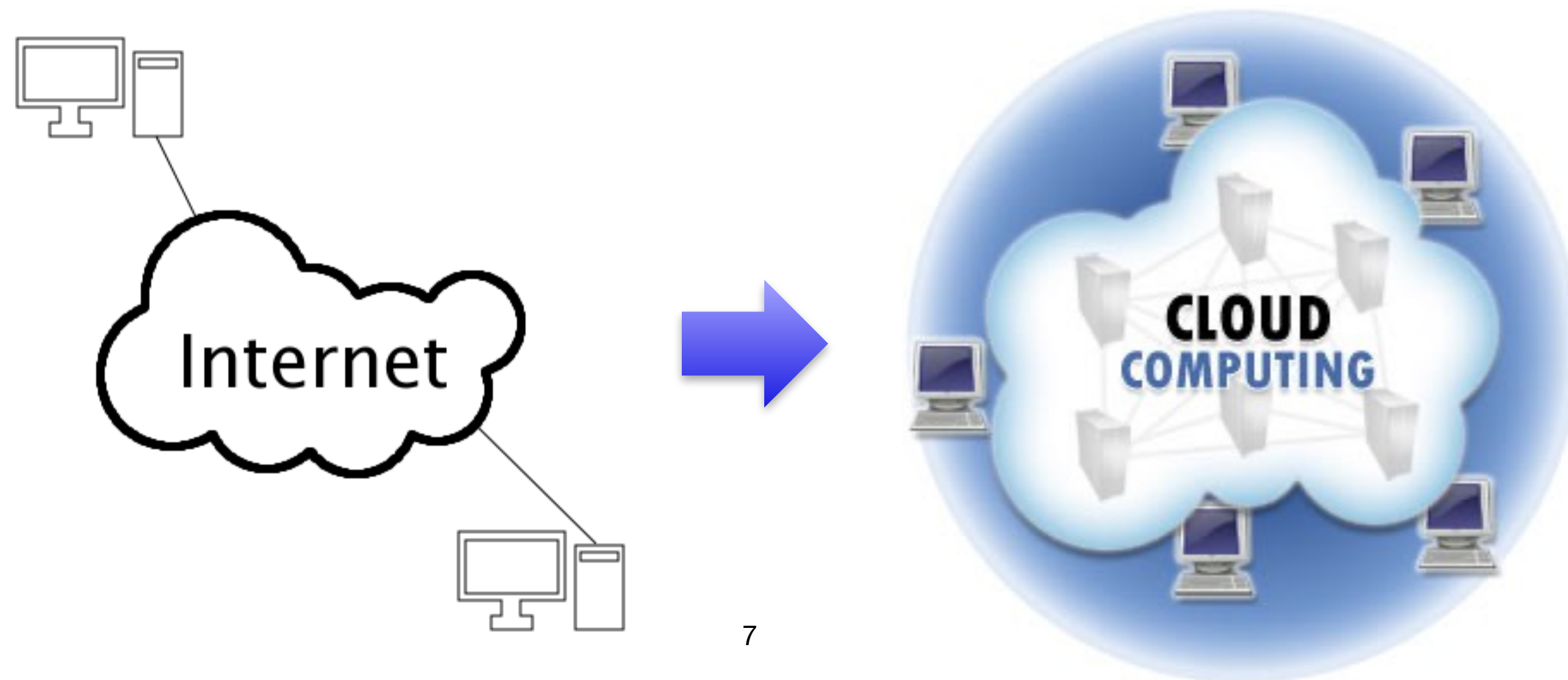
Cloud Computing refers to both the **applications delivered as services over the Internet** and the **hardware and systems software in the data centers that provide those services**.

Services have long been called Software as a Service (SaaS).

The hardware and software of the datacenter are what we will call a Cloud.

When a Cloud is made available to the public on a **pay-as-you-go basis**, the service being sold is **Utility Computing**.

Berkeley



What is Cloud Computing?

Cloud computing is a model for enabling **convenient, on-demand network access to a shared pool of configurable computing resources** (e.g., networks, servers, storage, applications, and services) that can be **rapidly provisioned and released** with minimal management effort or service provider interaction.

This cloud model promotes availability and comprises **five essential characteristics, three service models, and four deployment models.**



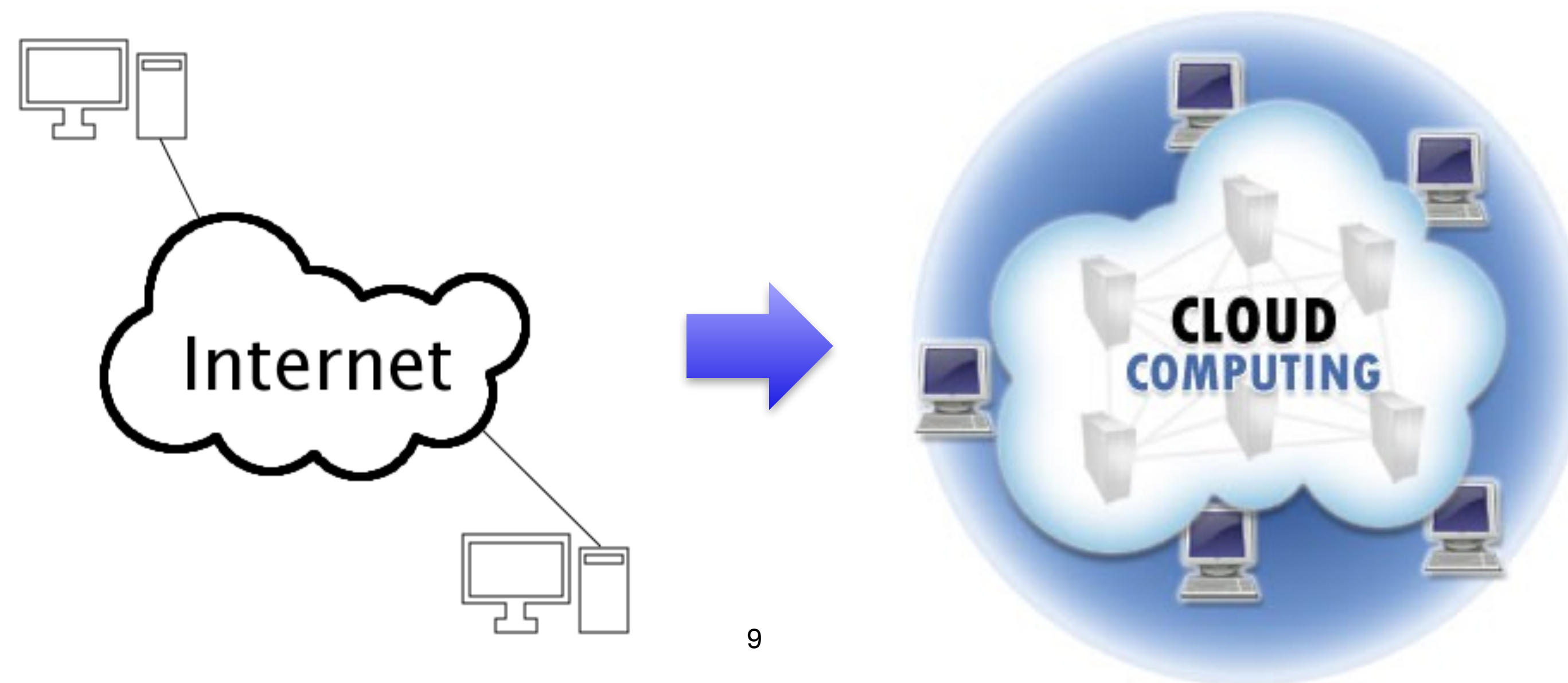
National Institute of Standards and Technology
Technology Administration, U.S. Department of Commerce

What is Cloud Computing?

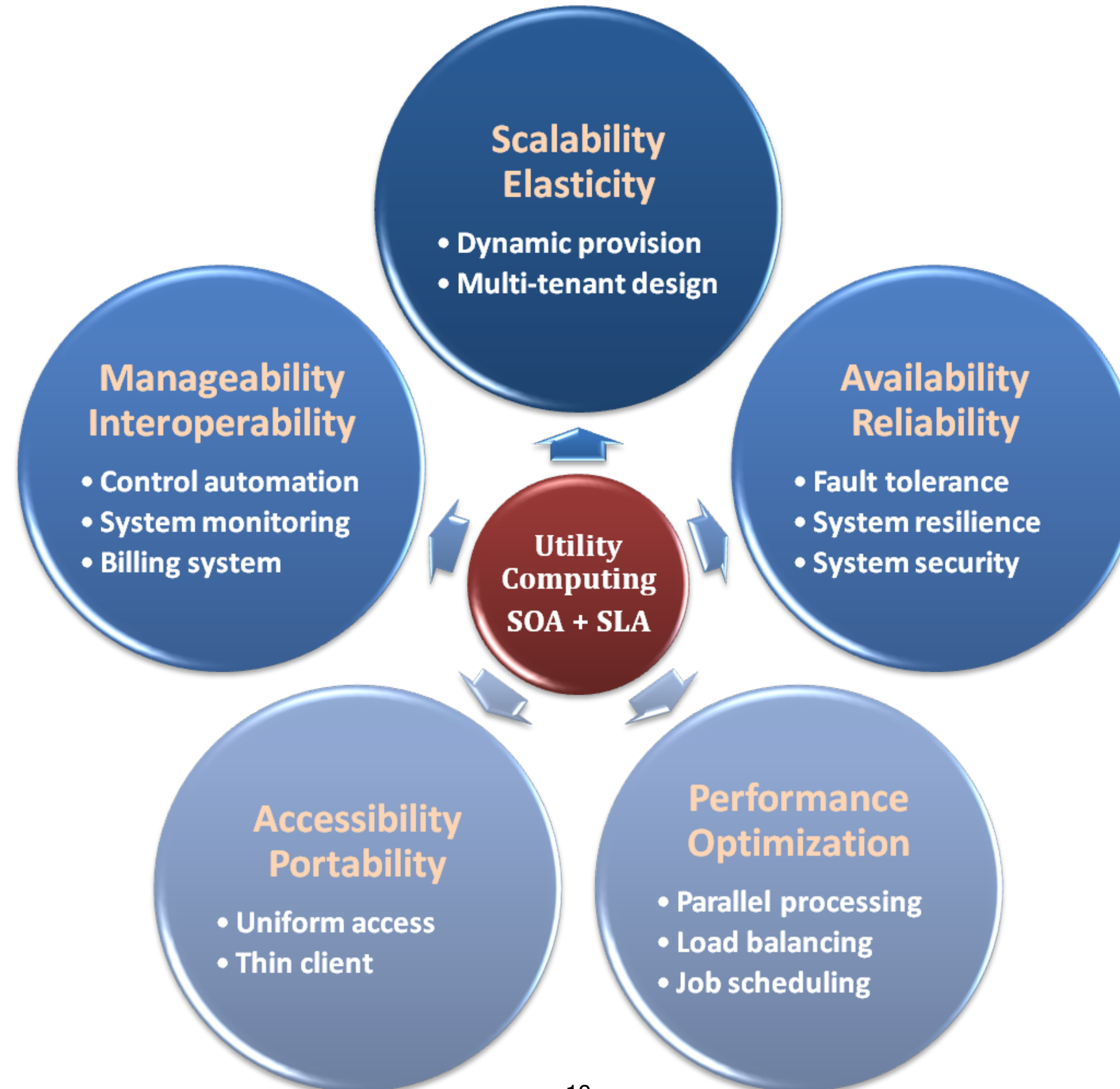
A Cloud is a distributed system consisting of **interconnected and virtualized computers** that are **dynamically provisioned and presented** as one or more unified computing resources based on **Service Level Agreements (SLA)** established through negotiation between the service provider and consumers.

In the Cloud, large groups of remote servers are networked to achieve coherence and economies of scale with the final goal of optimizing the effectiveness of the shared resources.

Cloud resources are not only **shared by multiple users** but are also **dynamically re-allocated on demand**.



The Five Cloud Characteristics



Services and Service-Oriented Architectures

A web service is a self-describing and stateless module that performs a work unit and is available over the network.
Web service providers offer APIs to exploit functionalities over the Internet instead of delivering full-stack applications.

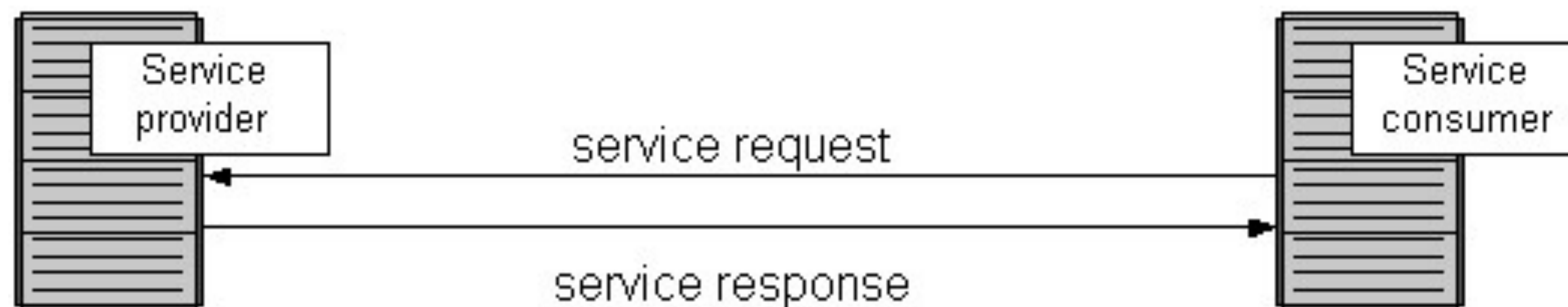
A service consists of a group of Web Services.

- A service should be well-defined.
- A service should be self-contained.
- A service should not depend on the context or state of other services.

A Service-Oriented Architecture (SOA) is a collection of services that communicate.

Contain a flexible set of design principles used during systems development and integration.

Provide a loosely-integrated suite of services that can be used within multiple business domains.



Quality Of Service and Service Level Agreements

Originally, Quality of Service (QoS) was a set of technologies for cost-effectively managing network traffic to enhance user experiences in home and enterprise environments.

Nowadays, QoS is a broad term used in the following areas:

- Customer care evaluations
- Technological evaluations

A service-level agreement (SLA) is a contract between a network service provider and a customer that specifies what services the service provider will provide, usually in measurable terms (QoS).

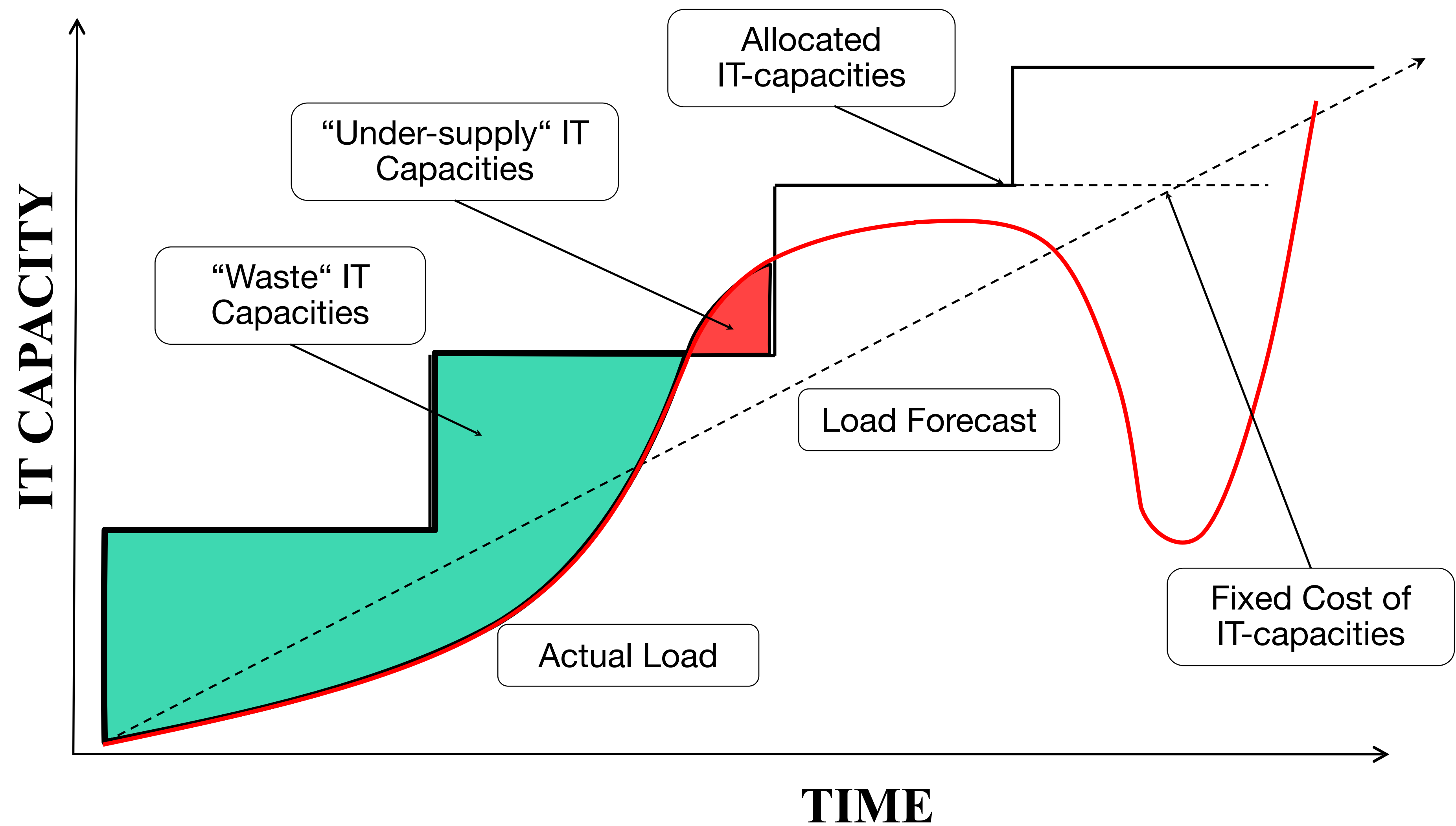
Typical content in contracts:

(i) performance guarantee metrics, (ii) up-time and down-time ratio, (iii) system throughput.

How to improve response time:

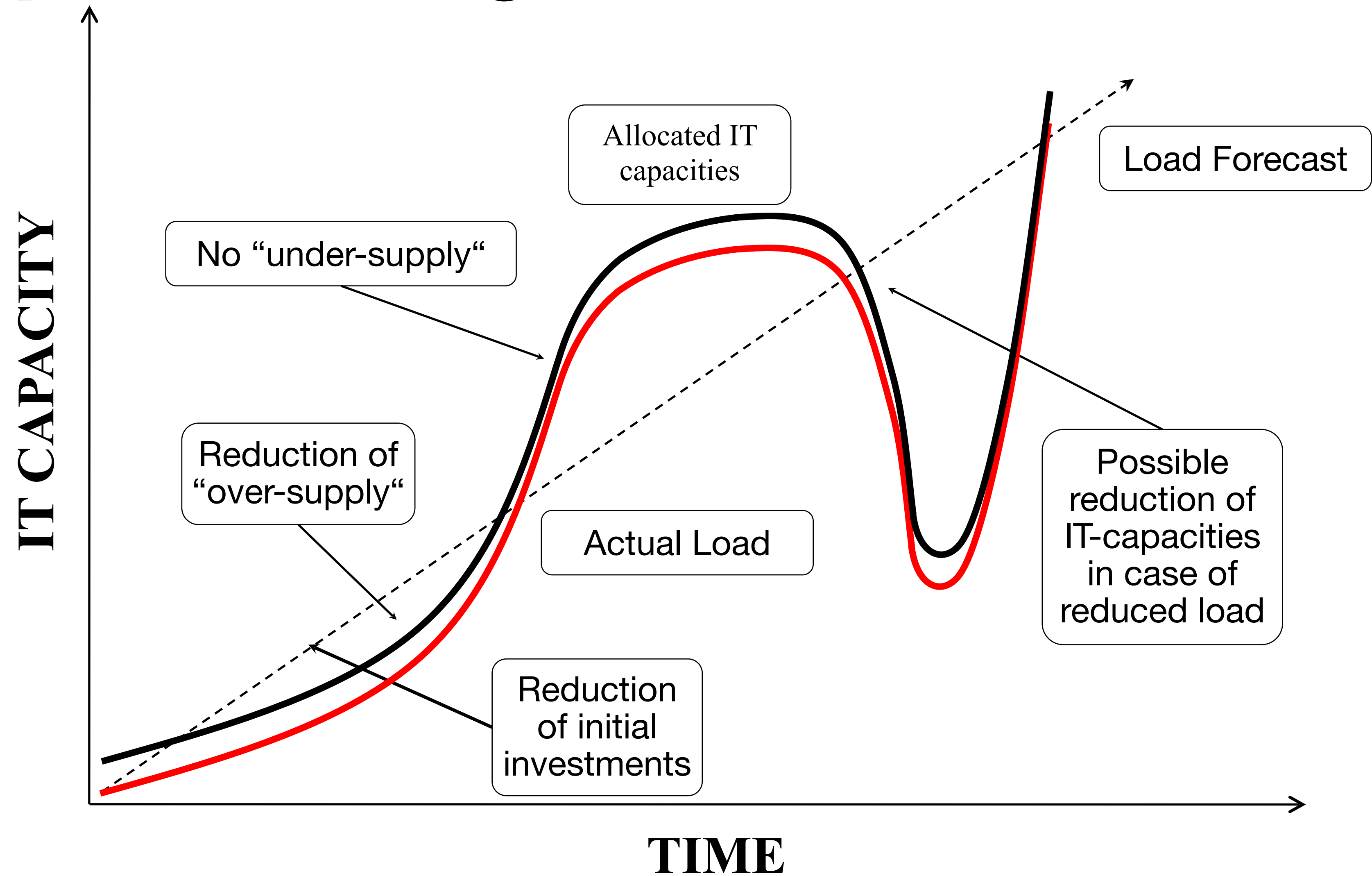
(i) problem management, (ii) penalties for non-performance issues, (iii) documented security capabilities.

Over-provisioning – Out of Cloud



Courtesy of Microsoft

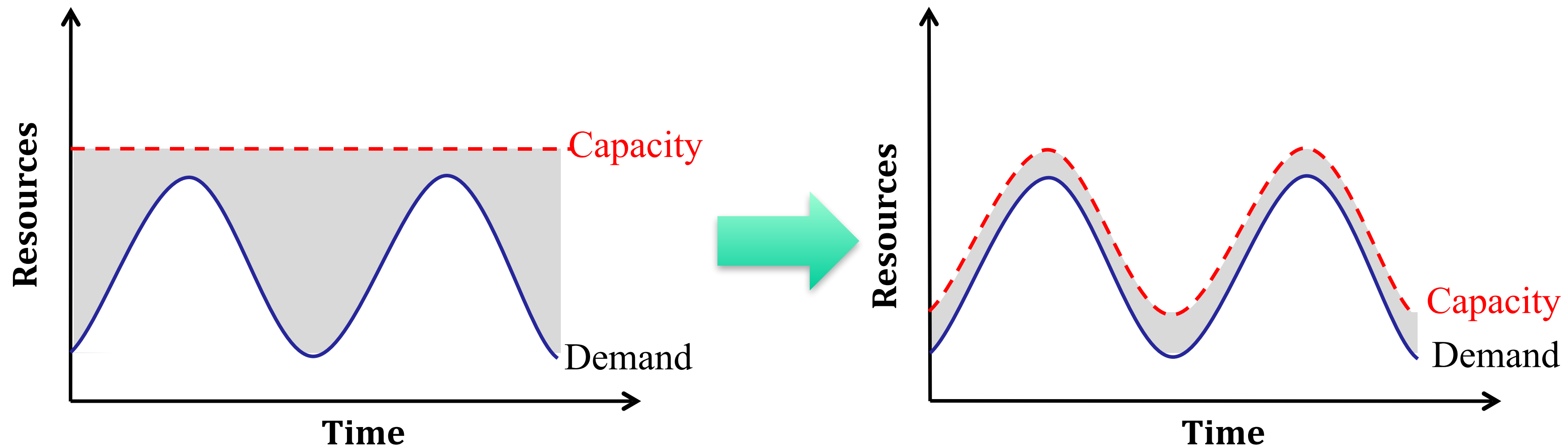
Cloud-provisioning



Dynamic Provisioning

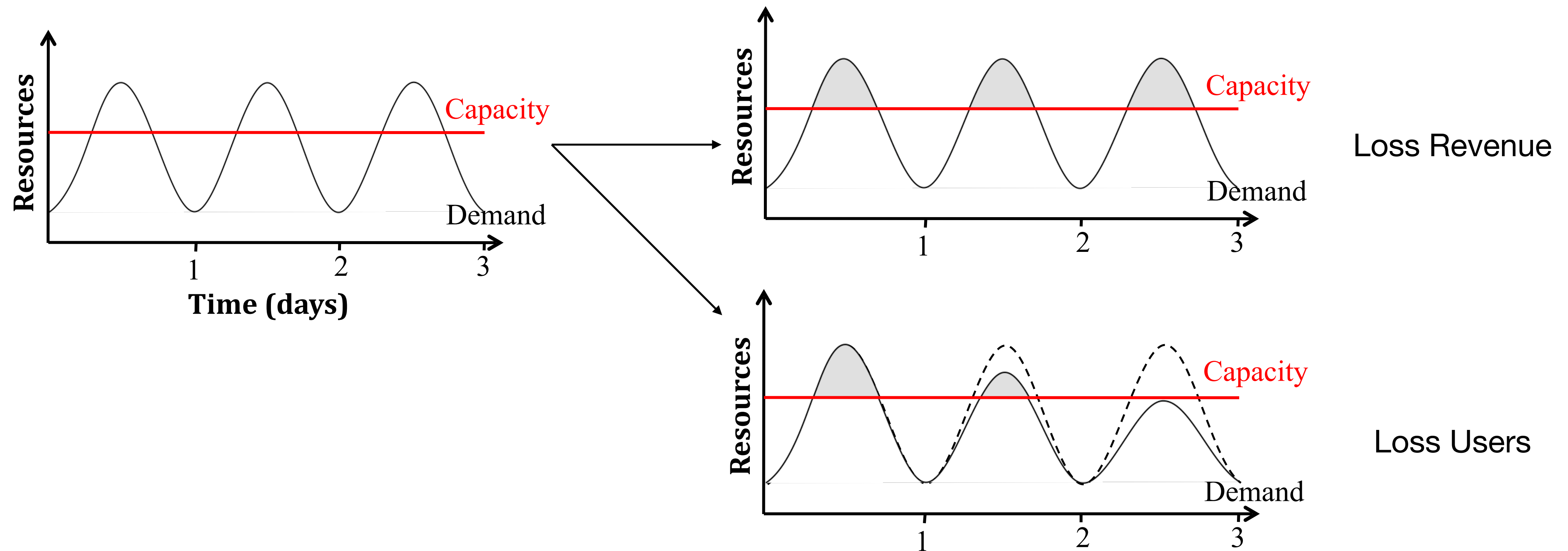
Cloud resources should be provisioned dynamically

- Meet seasonal demand variations
- Meet demand variations between different industries
- Meet burst demand for some extraordinary events



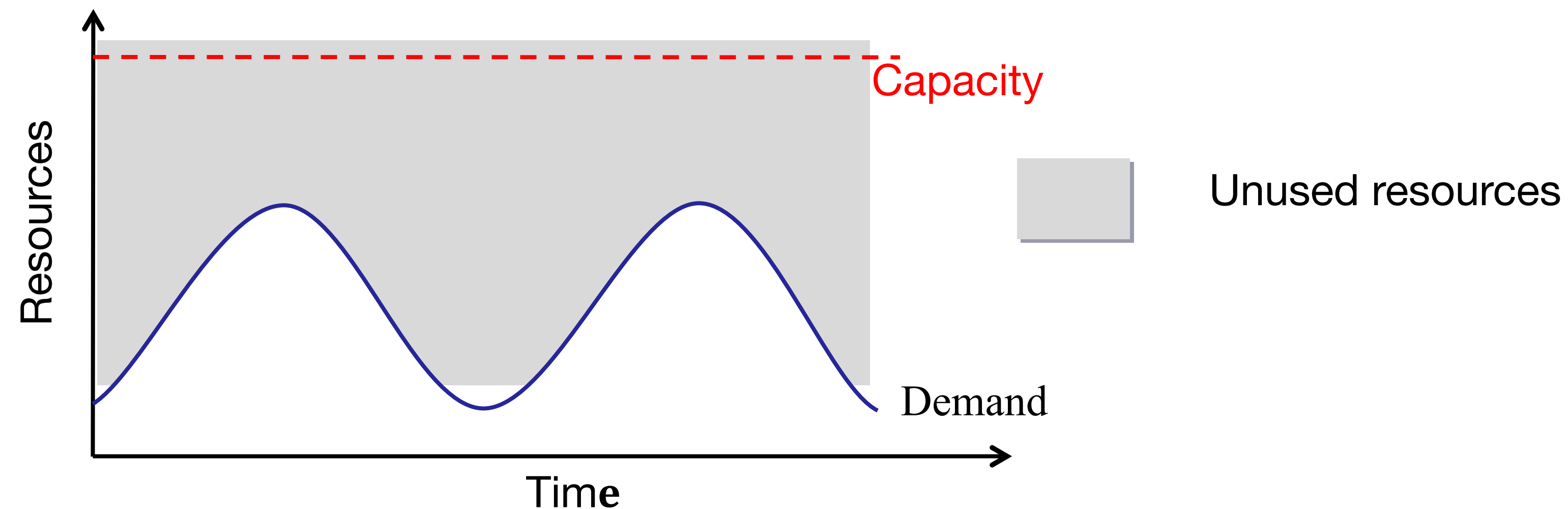
Dynamic Provisioning solves Under-provisioning

Under-provisioning: underestimate system utilization.



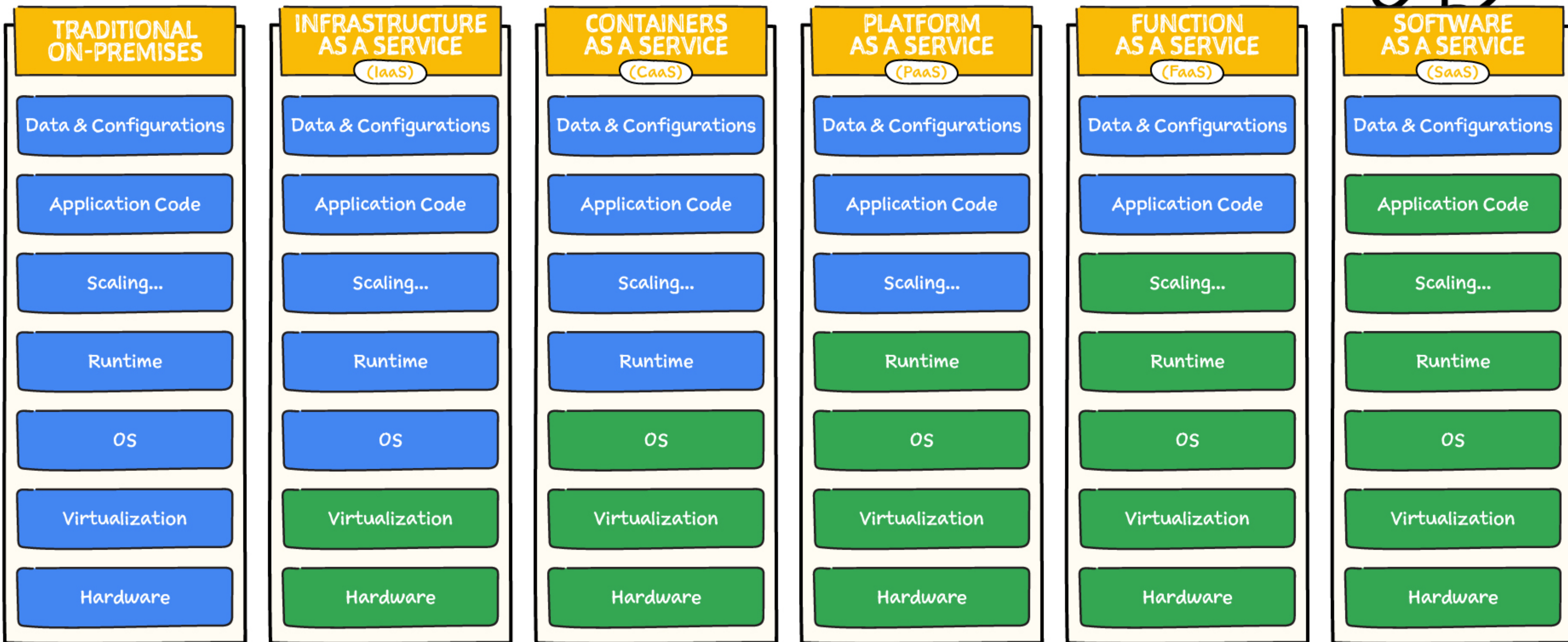
Dynamic Provisioning solves Over-provisioning

Over-provisioning: overestimate system utilization.





Wait... what is Cloud again?



You Manage

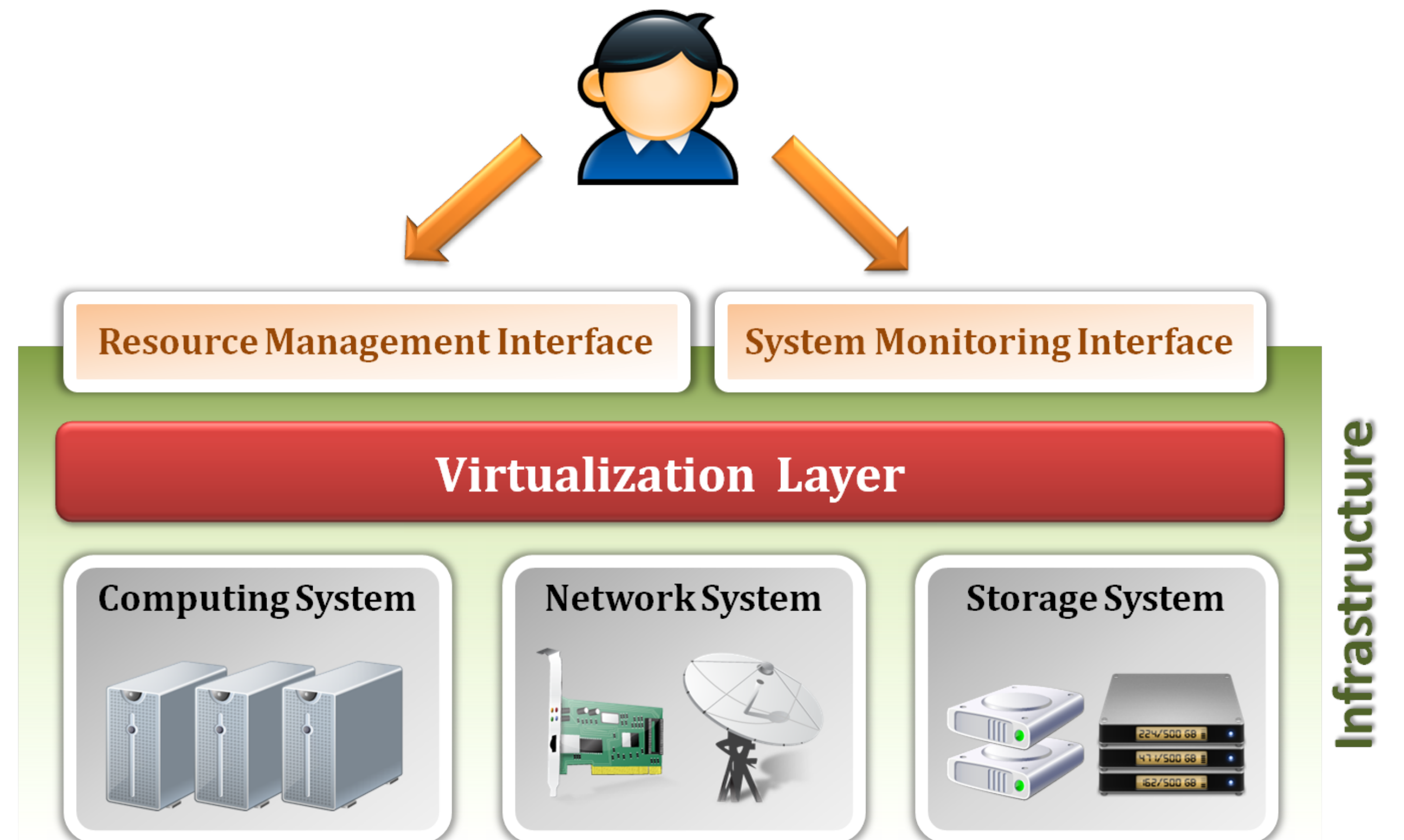


Cloud Provider Manages

Infrastructure as a Service (IaaS)

The capability provided to the consumer is to provision processing, storage, networks, and other computing resources where **the consumer can deploy and run arbitrary software**, including operating systems and applications.

The consumer does not manage or control the underlying cloud infrastructure but has **control over operating systems, storage, deployed applications, and possibly limited control of select networking components**.



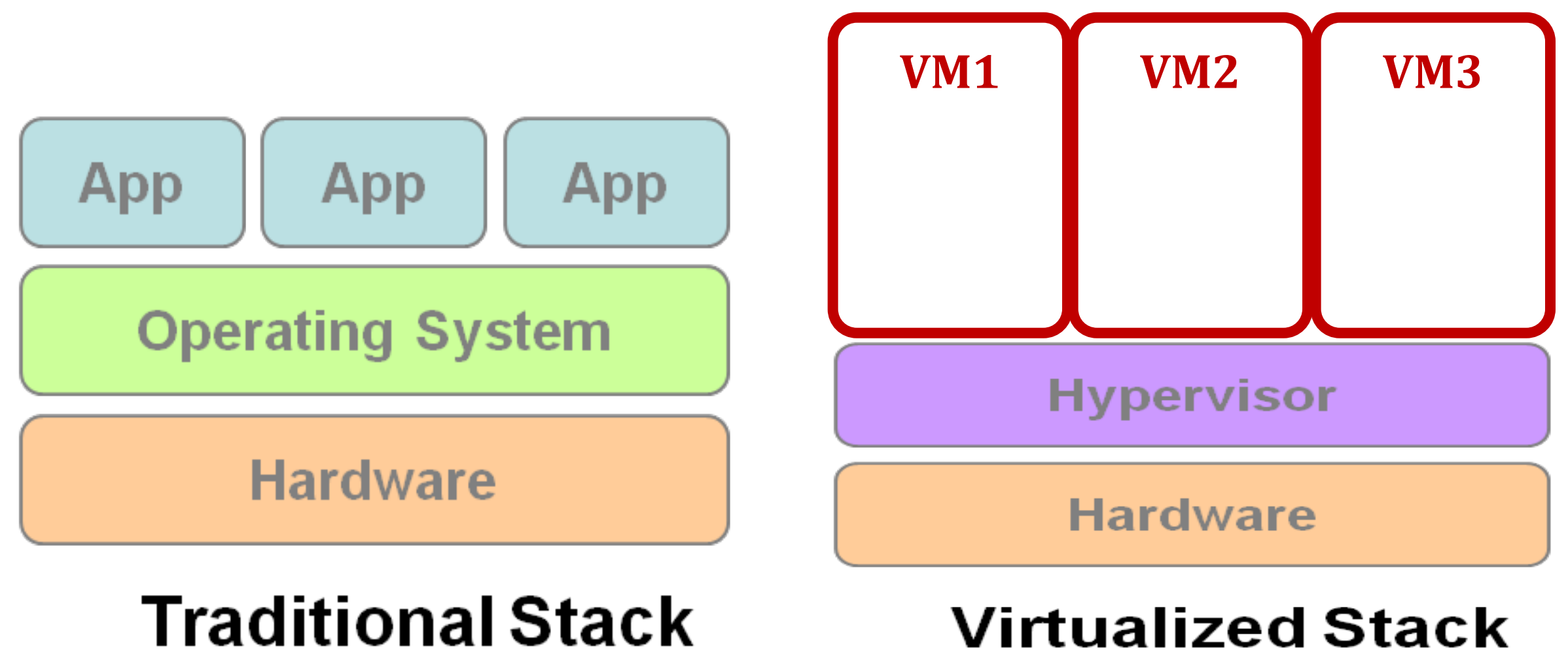
Infrastructure as a Service (IaaS)

Enabling technique -> Virtualization: an abstraction of logical resources away from underlying physical resources.

- Virtualization techniques shift OS onto hypervisor.
- Multiple OS share the physical hardware and provide different services.
- Improve utilization, availability, security, and convenience.

Provided Service: Resource Management Interfaces to provide the basic operations related to virtual machines, storage, and networks.

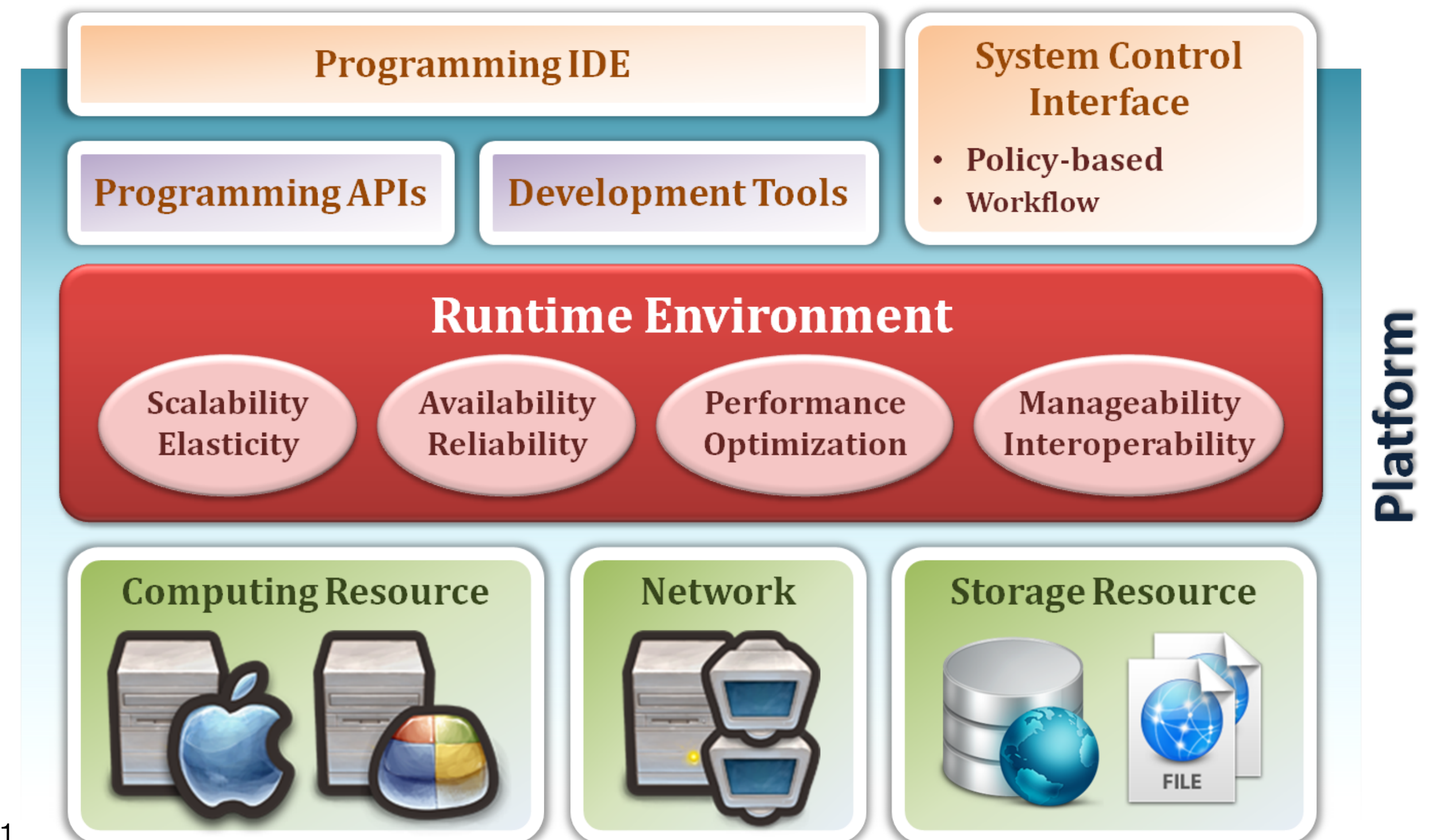
Provided Service: System Monitoring Interfaces monitor the system status of each virtual machine, virtual storage, and virtual network.



Platform as a Service (PaaS)

The consumer is provided with the capability to **deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages and tools supported by the provider.**

The consumer does not manage or control the underlying cloud infrastructure, including the network, servers, operating systems, or storage, but it does have **control over the deployed applications and possibly application hosting environment configurations.**



Platform as a Service (PaaS)

Enabling technique -> Runtime Environment Design: a collection of available software services usually implemented by program libraries.

Provided Service: Programming IDEs to develop their service among PaaS.

This IDE integrates the functionalities supported by the underlying runtime environment and provides development tools, such as a profiler, a debugger, and a testing environment.

The programming APIs supported from the runtime environment may vary between cloud providers, but some common operating functions still exist. They allow for operations related to computation, storage, and communication.

Provided Service: System Monitoring Interfaces through Policy-based and Workflow Control.

The former is a principle or rule to guide decisions and achieve rational outcomes(s).

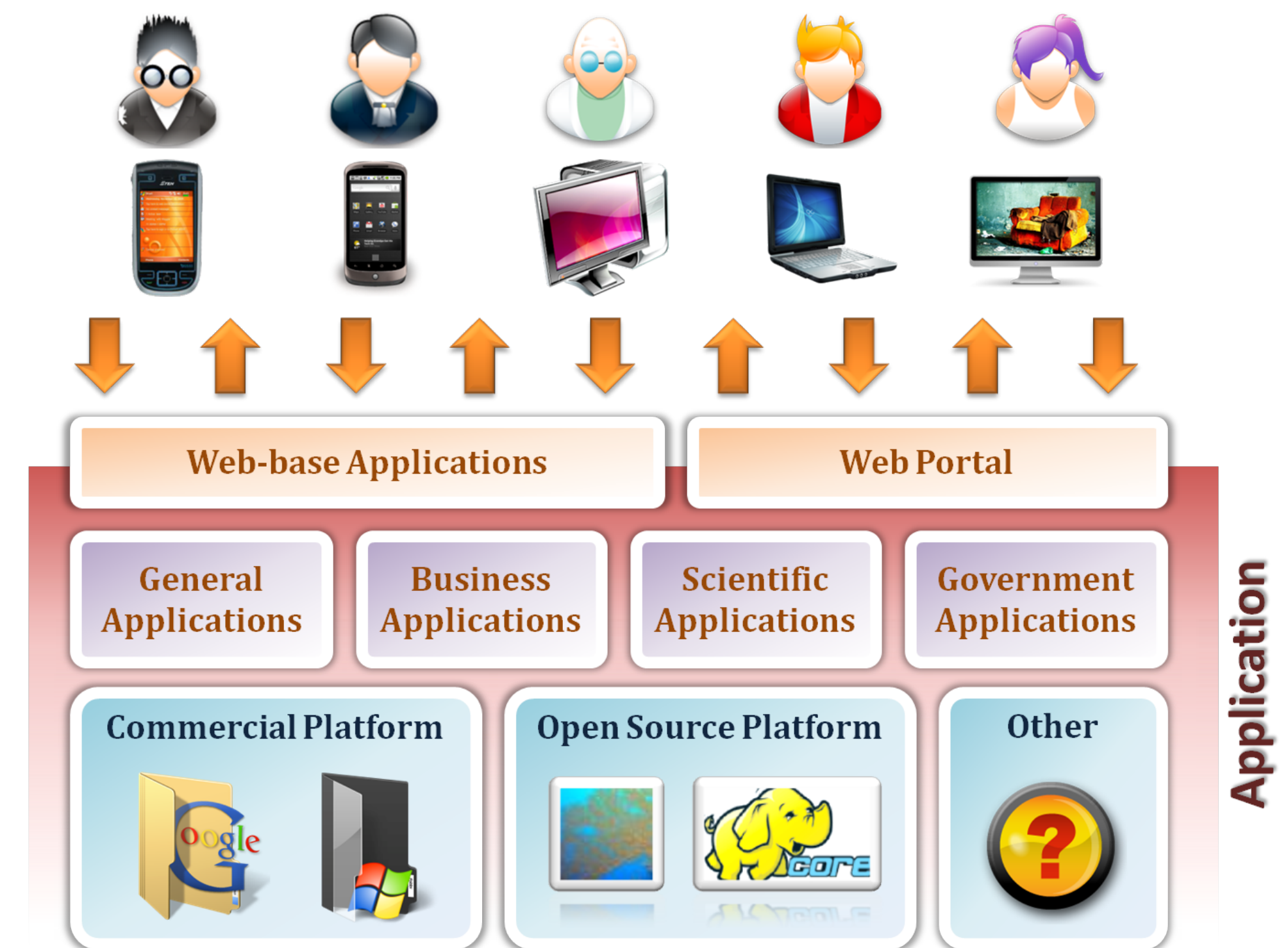
The latter describes the flow of installation and configuration of resources.

Software as a Service (SaaS)

The capability provided to the consumer is to **use the provider's applications running on a cloud infrastructure**.

The applications are accessible from various client devices through a thin client interface (e.g., web-based email).

The consumer does not manage or control the underlying cloud infrastructure, including network, servers, operating systems, storage, or even individual application capabilities, except for **limited user-specific application configuration settings**.



Software as a Service (SaaS)

Enabling technique -> Web Services

- Viewing the Internet as a computing platform
- Running interactive applications through a web browser
- Leveraging interconnectivity and mobility of devices
- Enhanced effectiveness with greater human participation

Provided Service: Web-based Applications

- General Applications - e.g., office suite, multimedia, and instant messages.
- Business Applications - e.g., ERP, CRM, and market trading system.
- Scientific Applications - e.g., aerospace simulation and biochemistry simulation.
- Government Applications - e.g., the national medical system and public transportation system services.

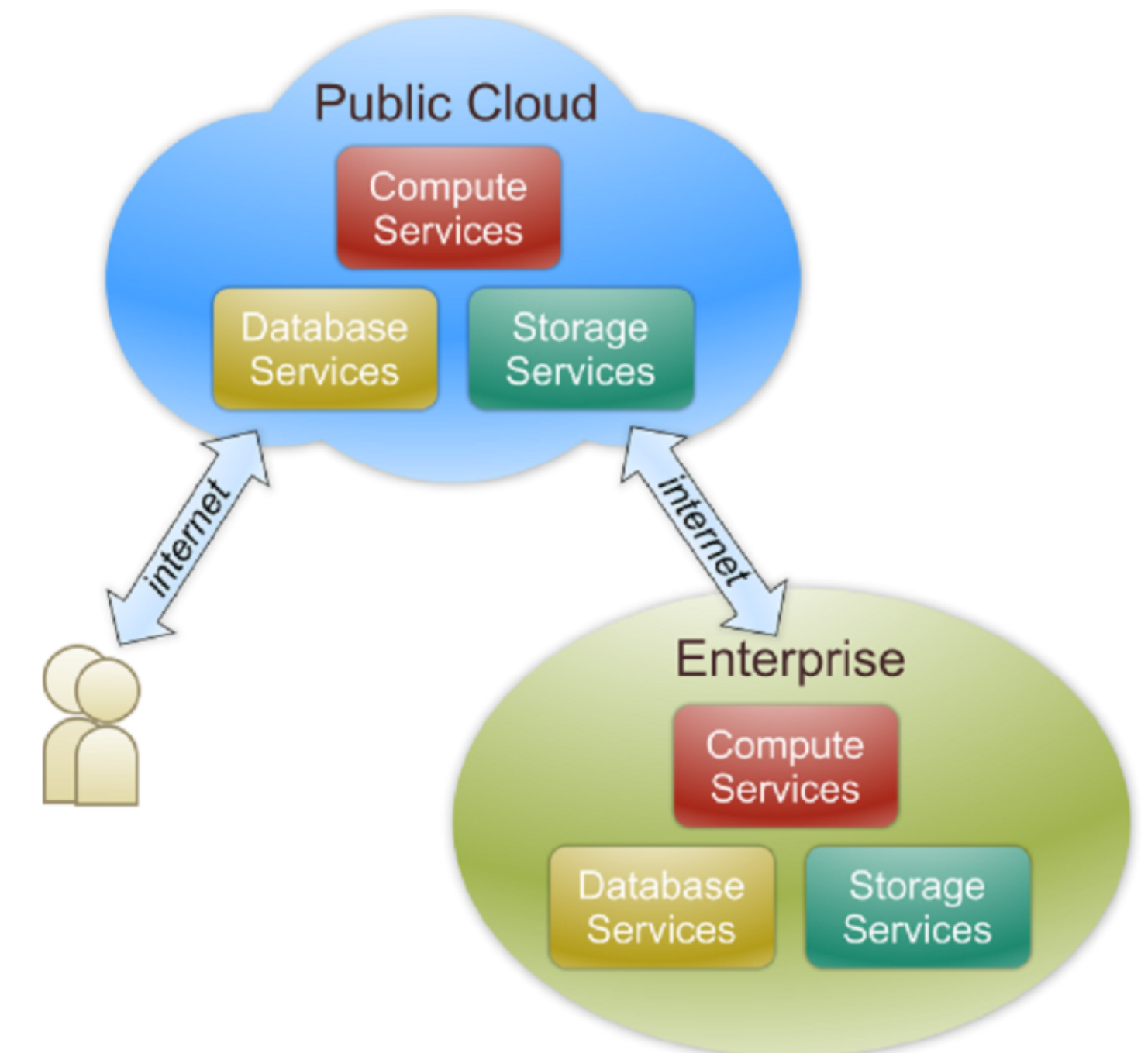
The Four Deployment Models: Public Cloud

The cloud infrastructure is made **available to the general public or a large industry group** and is owned by an organization selling cloud services.

Also known as external cloud or multi-tenant cloud, this model represents a cloud environment that is openly accessible.

Basic characteristics

- Homogeneous infrastructure
- Common policies
- Shared resources and multi-tenant
- Leased or rented infrastructure
- Economies of scale



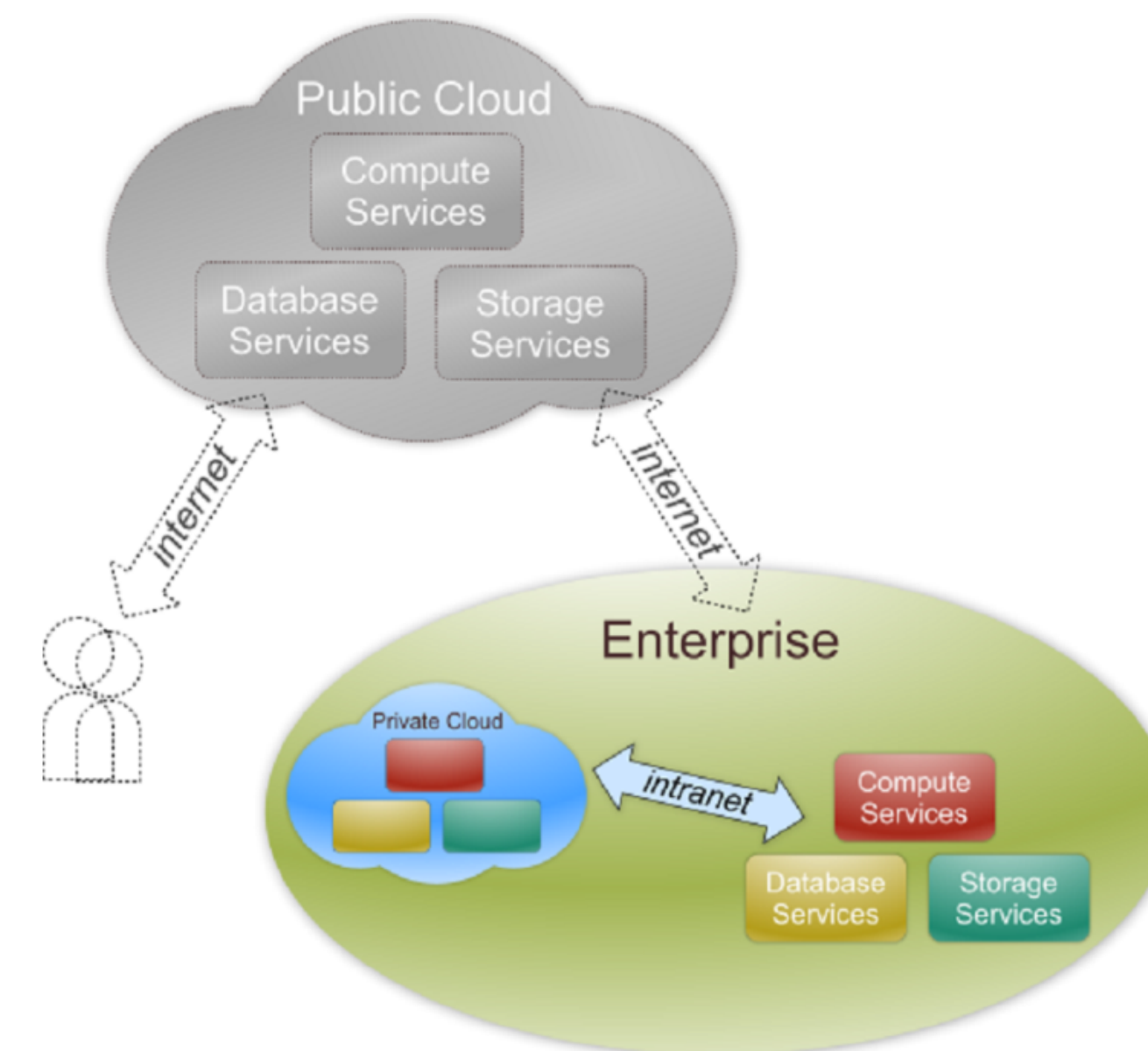
The Four Deployment Models: Private Cloud

The cloud infrastructure is **operated solely for an organization**. The organization or a third party may manage it. The cloud infrastructure may exist **on-premise** or **off-premise**.

Also known as an internal or on-premise cloud, a private cloud intentionally limits access to its resources to service consumers who belong to the same organization that owns the cloud.

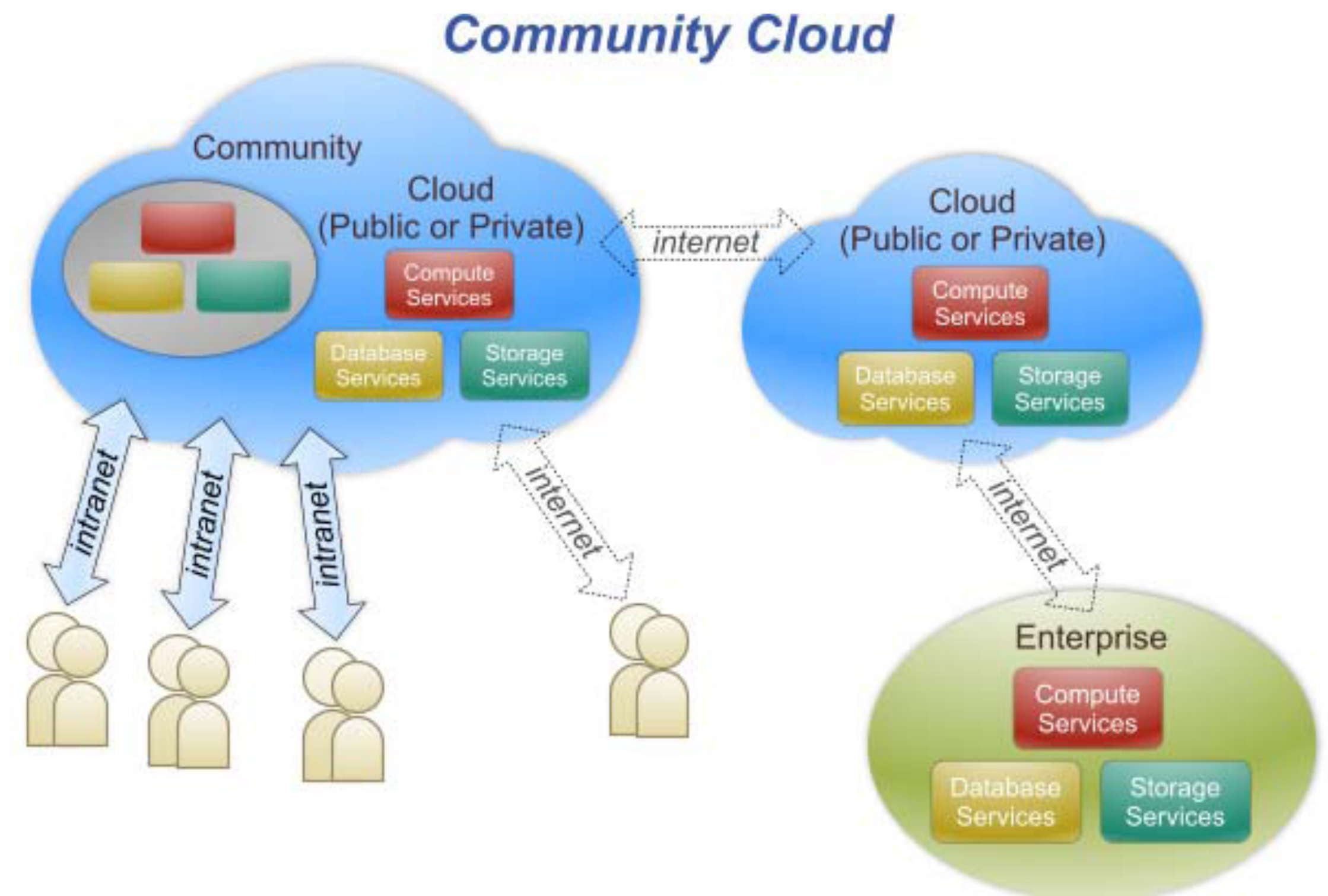
Basic characteristics

- Heterogeneous infrastructure
- Customized and tailored policies
- Dedicated resources
- In-house infrastructure
- End-to-end control



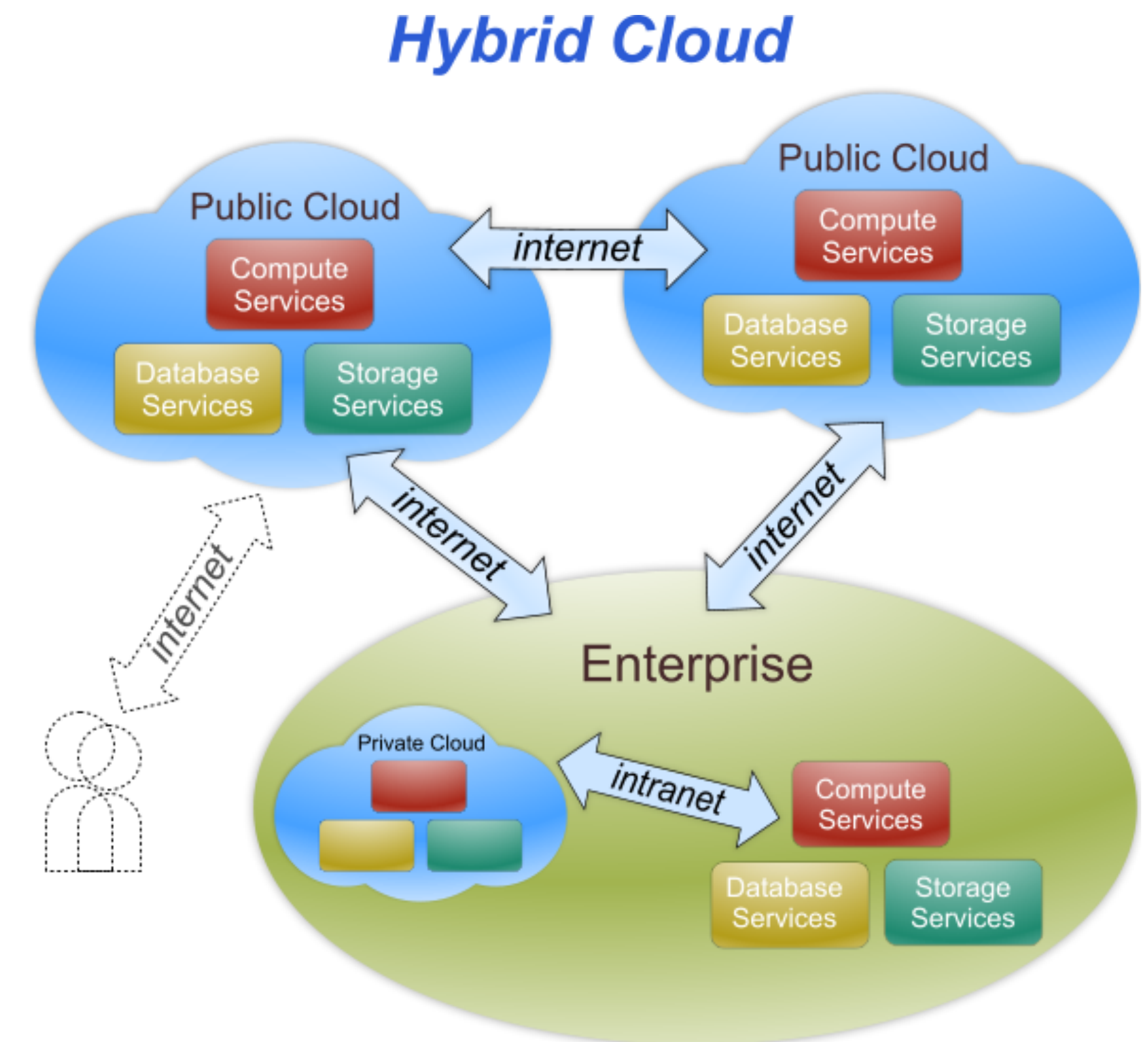
The Four Deployment Models: Community Cloud

Several organizations share the cloud infrastructure, which supports a specific community with **shared concerns** (e.g., mission, security requirements, policy, and compliance considerations).

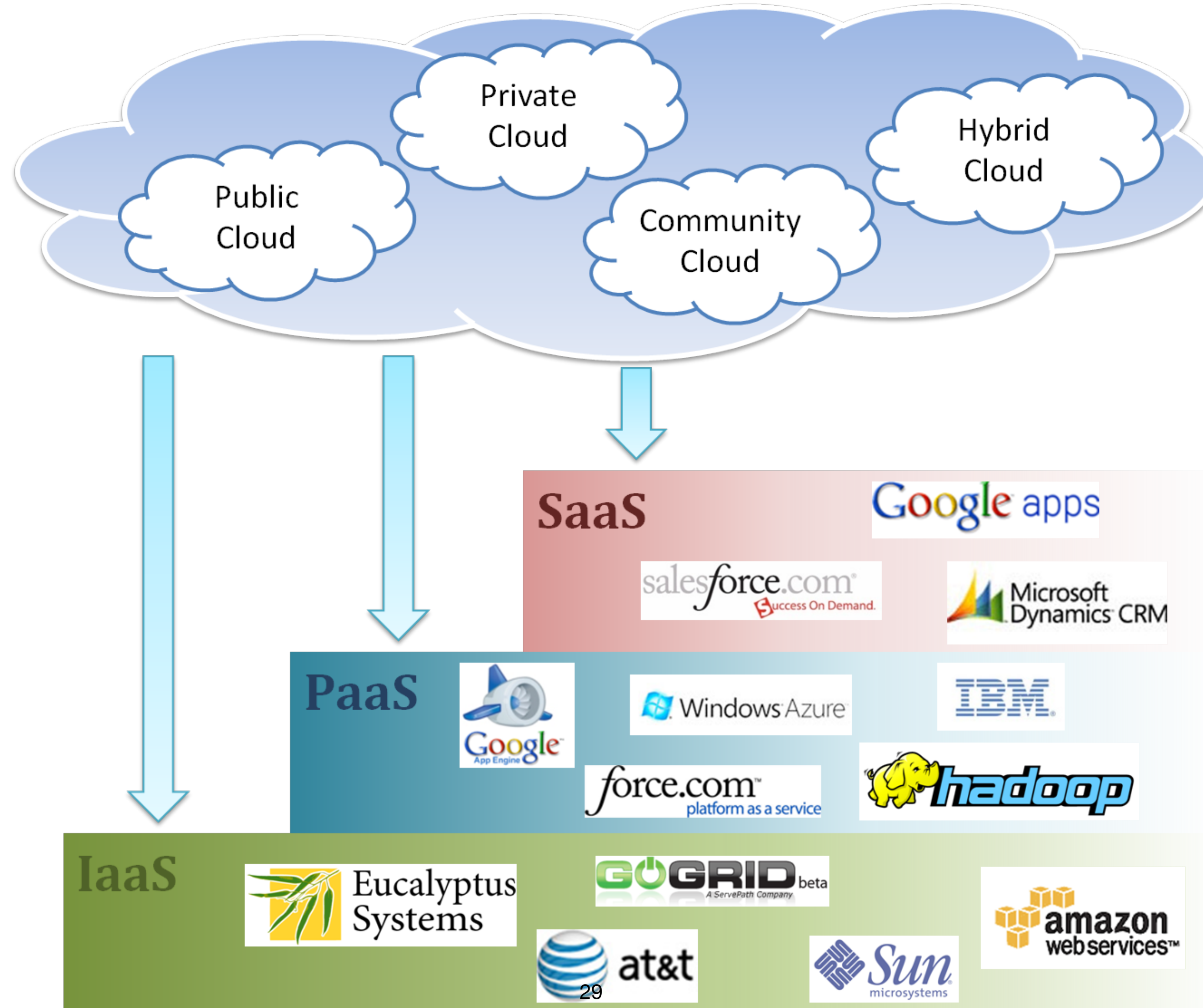


The Four Deployment Models: Hybrid Cloud

The cloud infrastructure is a **composition of two or more clouds** (i.e., private, community, or public) that remain unique entities but are **bound together by standardized or proprietary technology** that enables data and application portability (e.g., cloud bursting for load-balancing between clouds).



Cloud Ecosystem



A close-up photograph of a computer keyboard. The central focus is a large, white, rectangular key with rounded corners. On this key, there is a dark blue icon of a coffee cup with three wavy lines above it representing steam. Below the icon, the word "Break" is printed in a dark blue, serif font. The key is set against a light-colored, textured keyboard surface. Other keys are visible in the background, including one with a double quote symbol and another with a four-way arrow symbol, but they are out of focus.

Break